SECTION A: Answer all questions

QUESTION 1: (20 marks)

[a] **The following are multiple choice questions. Chose the correct answer from given alternatives for question (1) to (10) and write its letter in the answer sheet. (10 marks)**

1) What type of reaction do coordination complexes (complex ions) undergo in solution?

A. ligand exchange reactions
B. aromatic substitution reactions
C. acid-base neutralization reactions
D. oxidation-reduction reactions

2) The metal ion with the smaller atomic radii form stronger bonds hence;

A. complexes are less labile
B. complexes are extremely labile
C. complexes are inert
D. complexes are kinetically

3) According to Werner's explanations which of valency type determines the geometry of the complex?

A. secondary valency
B. primary valency
C. outer-shell valency electrons
D. inner-shell valency electrons

4) Werner's experimentations on $Co.Cl_3.5NH_3$ (purple) with silver observed the presence of:

A. 2 chloride ions
B. 3 chloride ions
C. 1 chloride ion
D. 0 chloride ions

5) The complexes $[CO(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$ are the examples of which type of isomerism?

A. Coordination isomerism
B. Ionization isomerism
C. Geometrical isomerism
D. Linkage isomerism

6) "Cobalt Yellow" is a pigment used in oil paints, and contains the coordination compound $K_3[Co(NO_2)_6]$. How many unpaired electrons are there on the cobalt atom in this compound? Note that NO_2^- is a strong-field ligand.

A. 0
B. 1
C. 2
D. 3

7) Which of the following is true about the complex $[PtCl_2(H_2O)(NH_3)]$?

A. It exhibits geometrical isomerism
B. It is paramagnetic complex
C. Platinum is sp^3 hybridised
D. It's geometry is tetrahedron

8) Consider the coordination compound, $K_2[Cu(CN)_4]$. A coordinate covalent bond exists between;

A. K^+ and CN^-
B. K^+ and $[Cu(CN)_4]^{2-}$
C. Cu^{2+} and CN^-
D. C and N in CN^-

9) The ultra violet and visible spectra of coordination compounds of transition metals involve transitions between;

A. The d ortbitals metals
B. The d orbitals non-metals
C. The s orbitals metals
D. both s and d orbitals metals

10) The absorbance or the optical density of a molecule absorbing in ultraviolet or visible region depends on:

A. molecular structure only
B. its density and volume only
C. its concentration
D. wavelength

11) Which one of the following statements is FALSE?

A. In an octahedral crystal field, the d electrons on a metal ion occupy the e_g set of orbitals before they occupy the t_{2g} set of orbitals.

B. Diamagnetic metal ions cannot have an odd number of electrons.

C. Low spin complexes can be paramagnetic.

D. In high spin octahedral complexes, oct is less than the electron pairing energy, and is relatively very small.

12) Consider the violet-coloured compound, $[Cr(OH_2)_6]Cl_3$ and the yellow compound, $[Cr(NH_3)_6]Cl_3$. Which of the following statements is false?

A. octahedral energy separation for $[Cr(NH_3)_6]^{3+}$ is calculated directly from the energy of yellow light.

B. Both chromium metal ions are paramagnetic with 3 unpaired electrons.

C. octahedral energy separation for $[Cr(OH_2)_6]^{3+}$ is less than oct for $[Cr(NH_3)_6]^{3+}$.

D. he two complexes absorb their complementary colours.

13) Which one of the following statements is FALSE?

A. In an octahedral crystal field, the d electrons on a metal ion occupy the e_g set of orbitals before they occupy the t_{2g} set of orbitals.

B. Diamagnetic metal ions cannot have an odd number of electrons.

C. Low spin complexes can be paramagnetic.

D. In high spin octahedral complexes, oct is less than the electron pairing energy, and is relatively very small.

14) The complex of $[Hg(CN)_4]^{2-}$ is thermodynamically stable as well as labile with the formation constant of 10^{42}, the reasons for this is.

A. When in solution it exchanges CN^- ligands with labelled cyanide ions, $^{14}CN^-$ at a very fast rate

B. When in solution it does not exchange CN^- ligands with labelled cyanide ions, $^{14}CN^-$ at a very fast rate.

C. When in solution it exchanges CN^- ligands with labelled cyanide ions, $^{14}CN^-$ at a slow rate.

D. All of the above

15) $[CoCl_2(NH_3)_4]^+$ + Cl^- → $[CoCl_3(NH_3)_3]$ + NH_3. If in this reaction two isomers of the product are obtained, which is true for the initial (reactant) complex?

A. compound is in *cis*-form
B. compound is in *trans*-form
C. compound is in both (*cis* and *trans*) form
D. None of the above

16) A complex ion contains a Cr^{+3} bonded to four H_2O molecules and two Cl^- ions. Write the formula of this complex.

A. $[Cr(H_2O)_4Cl_2]^{1+}$
B. $[Cr(H_2O)_4Cl_2]^{2+}$
C. $[Cr(H_2O)]^{1+}$ Cl^-
D. None of the above

17) Coordination number is a characteristic of which of the following?

A. Central atom
B. Ligand
C. Coordination entity
D. Coordination compound

18) Strong field ligands such as CN^- :

A. usually produce low spin complexes and small crystal field splitting.
B. usually produce high spin complexes and small crystal field splitting.
C. usually produce low spin complexes and high crystal field splitting.
D. usually produce low spin complexes and high crystal field splitting.

19) In coordination chemistry, the donor atom of a ligand is

A. the atom in the ligand that shares an electron with the metal.
B. the atom in the ligand that shares an electron pair with the metal.
C. the atom in the ligand that accepts a share in an electron pair from the metal.
D. the atom in the ligand that accepts a share in an electron pair from the counter ion.

20) Which of the following statement is true?

A. In *Mabcd* tetrahedral complex, optical isomerism cannot be observed.
B. In *Mabcd*, square planar complexes show both optical as well as geometrical isomerism.
C. In $[PtCl_2(NH_3)_2]^{2+}$ the *cis* form is optically inactive while trans form is optically active

D. In $[Fe(C_2O_4)_3]^{3-}$, geometrical isomerism does not exist while optical isomerism exists.

[b] **The following questions (21) to (30) are true or false statements. Correctly, circle "T" if the statement is correct and "F" if the statement is incorrect, in the answer sheet provided. (5 marks)**

21) The *cis*-$[Co(en)_2Cl_2]^+$ is the stereomeric optically active species.

22) If energy separation (Δ) is less than pairing energy electrons in the d orbital pairing occur before singly distributed into higher energy orbital, the e_g set.

23) The synthesis of cisplatin from the starting material $PtCl_4^{2-}$ in ammonia it is such that Cl^- ion plays a role as strong trans-effect for a second NH_3 to substitute a chloride ion trans-to the first NH_3.

24) Complexes $[Pt(OH)_2(NH_3)_4]SO_4$ and $[Pt\ (SO_4)(NH_3)_4](OH)_2$ are typically ionization isomers which can be distinguished by reacting with barium sulphate whereby the former forms white precipitates.

25) Dissociative mechanisms are more common for systems with high coordination numbers.

26) A complex formed by Cu^{2+} ion in reaction (1) is most stable than the one formed in reaction (2).

$[Cu^{2+} + 4\ CN^- \longrightarrow [CuCN_4]^{2-}$ logK = 27.3 ....(1)

$[Cu^{2+} + 4\ NH_3 \longrightarrow [Cu(NH_3)_4]^{2+}$ logK =11.6 ....(2)

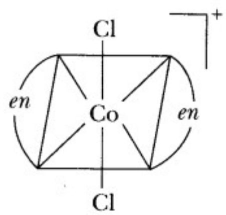
27) The SCN^- ion is a unidentate ligand.

28) The J-T distortion of the Cu^{2+} ion complex assumption d_{z^2} is singly filled results to elongation along the d_{z^2}.

29) One of the Werner's postulates is that when two different atoms with valence orbitals overlap with each other, it results in the formation of covalent bonds with the ligands.

30) The metal ion Cr^{2+} with d^4 configuration high spin its complexes are inert.

[c] **Matching items questions (31-35 and 36-40). Match the correct term from column B to suit items in column A. (5 marks)**

Column A	**Column B**
31) Proposed the chain theory to explain coordination compounds in the year 1869.	A. optically active B. optically inactive
32) The discoverer of coordination compound in year 1798.	C. $-2.7\Delta_{oct}$ D. $-0.53\ \Delta_{oct}$
33) Convert the splitting energy for tetrahedral $-1.2\Delta_{tet}$ to Δ_{oct}.	E. $-0.4\Delta_o + 1P$ F. $-1.6\ \Delta_o + 1P$
34) Given d^6 H.S electronic configuration, what will be the CFSE $_{(Oh)}$?	G. B.M Tassaert H. S. M. Jørgensen
35) 	I. A. Werner J. C. W.Blomstrand

Column A	**Column B**
36) Chlorophyll is considered a complex molecule due to presence of______ as central metal.	A. ethylenediamine B. ethylene
37) When 1 mol of $CrCl_3.6H_2O$ is treated with excess of $AgNO_3$, 3 mol of AgCl is obtained. The formula of the complex is _______	C. $[Cr(H_2O)_6]Cl_3$ D. Absorbs λ=red E. Absorbs λ=yellow F. $[Cr(H_2O)_3Cl_3]\cdot 3H_2O$
38) Example of a bidentate.	G. magnesium H. stabilize the t_{2g} level
39) Vacant high-energy π-orbital of Ligand	I. destabilize the t_{2g} level
40) The solution of complex ion $[Cr(H_2O)_6]^{3+}$ is violet.	J. ferrous

QUESTION 2: (20 marks)

[a] Write down the IUPAC name of the following complex: (4 marks)

(i) $[Pt(NH_3)(H_2O)Cl_2]$

(ii) $K_3[Fe(CN)_6]$

[b] Write the formula for the following complex: (4 marks)

(i) tris(ethane-l,2-diamine)chromium(III) chloride.

(ii) Pentaaminenitritocobalt(III) ion

[c] What is the difference between the following: (4 marks)

(i) Central metal and ligand

(ii) Charge of complex and oxidation state of metal ion

[d] List the postulates of Werner's theory. (4 marks)

[e] Illustrate the LFT approach for metal – ligand interaction with filled π ligands. (4 marks)

SECTION B: Answer <u>at least</u> ONE question from this section

QUESTION 3: (20 marks)

[a] Exchange of H_2O ligand on $[(CO)_3Mn(H_2O)_3]^+$ is much more rapid than on the analogous $[(CO)_3Re(H_2O)_3]^+$. The activation volume (change in volume on formation of the activated complex) is -4.5 ± 0.4 $cm^3\ mol^{-1}$.

(i) Suggest why the Mn complex reacts more rapidly than the analogous Re complex? (4 marks)

(ii) What is the difference between the interchange I_a and I_d mechanisms? (4 marks)

(iii) Is the activation volume more consistent with an A (or I_a) or a D (or I_d) mechanism? Explain. (2 marks).

[b] Strong π - acceptor ligands favour late transtion metals in low oxidation states.

(i) Draw a complete MO diagram for an octahedral complex having only σ -bonding interactions, such as $[Fe(H_2O)_6]^{2+}$. (6 marks)

(ii) How does replacing the water ligand with CO ligand generate a larger Δ_o? Use LFT to illustrate. (4 marks)

QUESTION 4: (20 Marks)

The complex $[Ti(H_2O)_6]^{3+}$ has its electronic spectrum given in Figure 1, which shows only one wavelength max. Study the complex and answer the following questions

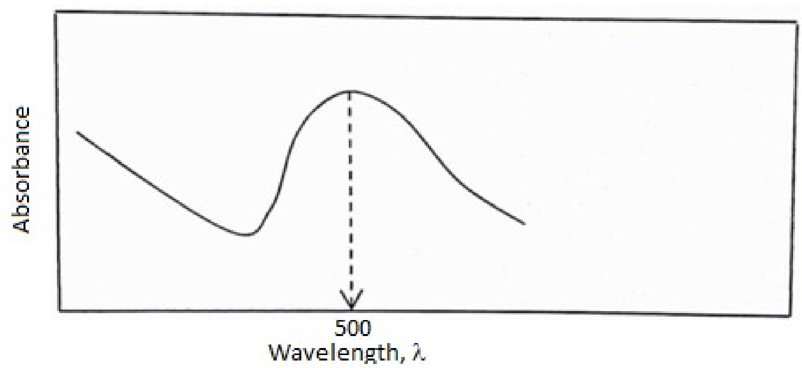


Figure 1

[a] How many transitions are taking place in this complex, describe why? What does that entail in terms of d orbitals splitting? (6 marks)

[b] Express the value of the crystal field splitting parameter (Δ) for this ion in cm^{-1} (2marks)

[c] Calculate the crystal field splitting energy (Δ) of this complex compound from its absorption spectrum. (6 marks)

[d] Plot the energy difference between t_{2g} and e_g sets of orbitals for the calculate energy above in (c). (4 marks)

SECTION C: Answer <u>at least</u> ONE question from this section

QUESTION 5: (20 marks)

[a] Solvent plays an important role in complex reaction and formation. Using water as a polar solvent explain how does it affect the thermodynamic and kinetic stability of the complex reaction. (4 marks)

[b] By mentioning at least one example in each case, describe four types of electronic spectra of complexes. (8 marks)

[c] When electrons are raised from lower to higher energy level, usually, colours are produced which relates to the magnitude of the spacing between the levels. Describe the four factors into which the spacing of the orbitals depends on: (8 marks)

Hints (geometry of complex, nature of ligand present, oxidation state of the central metal atoms, electronic spectra of complexes)

QUESTION 6: (20 marks)

[a] Describe the Jahn-Teller Effect, both elongation and compression along the d_{z^2} . (8 marks)

[b] With typical reactions explain the trans-effect in the *trans*- and *cis*- platin synthesis. (6 marks)

[c] Thermodynamic stability of the complex does not necessarily mean kinetically stability. Discuss. (6 marks)

Instructions from the cover page are emphasized.

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